MARK SCHEME · TEACHER COPY

12BR · Topic 18: Artificial Intelligence

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Total marks: 44

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · Oct/Nov 2025 Paper 31, original Q10 (a),(b) [6 marks]

Question	Answer	Marks
1(a)	**One** mark for each mark point (**Max 2**) MP1 A graph is used in AI to record relationships between entities MP2 … using vertices / nodes and edges MP3 for example, to represent places on a map and the distances between them, in order to find the shortest route.	2
1(b)	**One** mark for each mark point (**Max 4**) MP1 Artificial neural networks are designed to work in the same way as the human brain MP2 ANNs provide the architecture and algorithms for learning from the data MP3 They have a large number of connected processing units / nodes MP4 … that are arranged in layers / interconnected and work together to process data MP5 Deep learning models learn from data by adjusting the weights/biases of the connections between neurons MP6 They use multiple hidden layers to extract complex features and to make predictions	4

Question 2 · May/June 2026 Paper 33, original Q9 (a),(b) [8 marks]

Question	Answer	Marks	Guidance
2(a)	**One** mark per mark point (**Max 3**) MP1 A supervised learning technique of machine learning MP2 It finds a mathematical function // It plots a best-fit line/curve // calculus is used to find the error gradient MP3 … that maps how independent variables/cause influence a dependent variable/effect between the data MP4 It is used for forecasting trends and predicting outcomes MP5 It will be trained to understand the relationship between different causes and effects allow to find relationships between two **sets of data** MP6 The goal is to minimise the difference between the predicted values and the actual values.	3	
2(b)	**One** mark for each correct calculation as follows (**Max 4**) MP1 W to N2 **and** W to N3 (Lines 3 and 4 in table**)** MP2 W ⇨ N2 to N5 (Line 5 in table) MP3 W ⇨ N2 ⇨ N5 to N4 (Line 6 in table) MP4 W ⇨ N2 ⇨ N5 to N6 (Line 7 in table) MP5 W ⇨ N2 ⇨ N5 to E (Line 8 in table) MP6 W ⇨ N2 ⇨ N5 ⇨ N6 to E (Line 9 in table) **One** mark for correct path (**Max 1**) MP7 W ⇨ N2 ⇨ N5 ⇨ N6 ⇨ E	5	Lines 3 and 4 can be in either order. Lines 6, 7 and 8 can be in any order. Everything else must be in the table order. Shaded parts given in question.

Total: 44 marks

	Current path	Destination node	Cost of current path (g)	Heuristic (h)	Total (f = g + h)
2	W⇨N2⇨N5⇨ N6	E	4+7+4+3=18	0	18

Final path	W⇨N2⇨N5⇨N6⇨E

Question 3 · May/June 2026 Paper 32, original Q9 (a),(b) [9 marks]

Question	Answer	Marks	Guidance
3(a)	**One** mark per correct machine learning type (**Max 1**) Deep learning	**1**	Machine learning is NE because not all machine learning uses ANNs.
3(b)(i)	**One** mark per mark point (**Max 3**) MP1 The A* algorithm tries to find the shortest path using heuristics / intelligent guesses MP2 The A* algorithm doesn't necessarily pass through all nodes/paths MP3 Dijkstra's algorithm finds the shortest path to **all nodes** // Dijkstra's algorithm simply checks **each vertex** looking for the shortest path MP4 Dijkstra's algorithm **always** finds the shortest path MP5 The A* algorithm may not find the shortest path if the heuristics are inaccurate.	**3**	Do not award MP3 for searching through all **routes/paths** – it is looking at the shortest path to all **nodes**

3(b)(ii)	**One** mark per mark point - working (**Max 3**) May be seen on diagram or in working section MP1 Initialisation – setting Home to 0 MP2 … and the rest of the nodes to ∞ MP3 Evidence to show values at nodes **being updated** MP4 Evidence to show 'visited node(s)' MP5 Evidence to **show a correct calculation** of at least one route MP6 Evidence to show more than one route has been calculated for at least one node Correct Answers (**Max 2**) - allow these marks if totals clearly identified in working **Two** marks for all six correct values **One** mark for four or five correct values A: 6, B: 13, C: 9, D: 20, E: 17, F: 14	**5**	MP1 a 0 on the home node in the diagram or clearly labelled in working MP2 requires an ∞ symbol on EVERY node except Home in the diagram or clearly labelled in working MP3 evidence of correct numbers being written and amended on the diagram at the nodes to represent 'distance' for a particular route, or in the working MP4 any evidence of nodes being 'visited' in a route on the diagram OR listing of nodes by letter in the working MP5 numbers on the diagram or in working that show a route being calculated from the component distances between nodes MP6 numbers on the diagram or in working to show distances calculated to a node via different routes.

Question 4 · Oct/Nov 2025 Paper 32, original Q6 (a),(b) [6 marks]

Question	Answer	Marks
4(a)	**One** mark per point (**Max 2**) MP1 A graph uses vertices/nodes to identify/represent entities such as destinations, people, etc MP2 Edges are used to connect nodes and can represent possible paths between them // a path is the list of nodes connected by edges between two given nodes MP3 Nodes/edges can be labelled/weighted, and this is a weighting that can be applied and used in the context of the application MP4 A cycle is a list of nodes that return to the same node.	**2**

4(b)	**One** mark per mark point (**Max 4**) MP1 Supervised learning uses labelled data // Unsupervised learning makes use of unlabelled data. MP2 Labelled data means that known outcomes are applied to specific inputs to help the AI predict outcomes. MP3 Supervised learning requires initial human input/training // Unsupervised learning does not require human input/training. MP4 With unlabelled data in unsupervised learning, outcomes are not known MP5 … the AI has to search for hidden patterns/structures/clusters MP6 … within the data in order to predict outcomes.	**4**

Question 5 · May/June 2026 Paper 31, original Q9 (a),(b) [9 marks]

Question	Answer	Marks	Guidance
5(a)	**One** mark per mark point (**Max 4**) MP1 Supervised learning makes use of **labelled** training data MP2 … to predict future events based on past outcomes MP3 Unsupervised learning makes use of **unlabelled** training data / training data without instructions on what to do with it / does **not** use training data with the correct/expected outcomes MP4 … to search for hidden patterns/relationships/structures/similarities/ outliers within the data MP5 … to therefore get closer to the desired outcome as the training progresses.	**4**	

5(b)	**One** mark per mark point - working (**Max 3**) May be seen on diagram or in working section MP1 Initialisation – setting Begin to 0 MP2 … and the rest of the nodes to ∞ MP3 Evidence to show values at nodes being updated MP4 Evidence to show 'visited node(s)' MP5 Evidence to **show a correct calculation** of at least one route MP6 Evidence to show more than one route has been calculated for at least one node Correct Answers (**Max 2**) **Two** marks for all six correct values **One** mark for four or five correct values. P: 8; Q: 13; R: 6; S: 14; T: 17; U: 20	**5**	MP1 a 0 on the Begin node in the diagram or clearly labelled in working MP2 requires an ∞ symbol on EVERY node except Begin in the diagram or clearly labelled in working MP3 evidence of correct numbers being written on the diagram at the nodes to represent 'distance' for a particular route MP4 any evidence of nodes being 'visited' in a route on the diagram OR listing of nodes by letter in the working MP5 numbers on the diagram or in working that show a route being calculated from the component distances between nodes MP6 numbers on the diagram or in working to show distances calculated to a node via different routes.

Question 6 · Oct/Nov 2025 Paper 33, original Q10 (a),(b),(c) [6 marks]

Question	Answer	Marks
6(a)	To find the path between two points on a graph using the algorithm.	**1**
6(b)	**One** mark per point (**Max 2**) MP1 A* tries to find a better path (between two points) by using a heuristic function // A* finds the adjacent route with the shortest path and continues this until the destination is reached MP2 … Dijkstra's just explores all possible routes. MP3 The heuristic function on the A* algorithm gives priority to nodes that are supposed to be better than others / less costly than others. MP4 Dijkstra's algorithm cannot work with negative values/weights //A* algorithm can work with negative values/weights.	**2**
6(c)	**One** mark for each mark point (**Max 3**) MP1 Unsupervised learning uses algorithms to analyse / cluster MP2 … unlabelled data sets MP3 They discover hidden patterns / data groupings / clusters without the need for human intervention. MP4 It is able to discover similarities and differences in data / information.	**3**